C10 Using Resources – Knowledge Organiser

AQA GCSE Chemistry • Set 1: Sustainability and Water **Paper 2**

#	Question	Answer
1	Sustainable development	Meets present needs without harming future generations
2	Finite resource	Will run out
3	Renewable resource	Can be replaced
4	Potable water	Water safe to drink
5	Why potable water is not pure	Contains dissolved substances
6	Two stages to treat fresh water	Filtration, sterilisation
7	Sterilising agents	Chlorine, ozone, UV light
8	Methods of desalination	Distillation, reverse osmosis
9	Why desalination is expensive	Needs large amounts of energy
10	Waste water treatment stages	Screening, sedimentation, anaerobic digestion of sludge, aerobic treatment of effluent

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AQA GCSE Chemistry • Set 2: Metal Extraction and Life Cycle Assessment **Paper 2**

#	Question	Answer
1	Phytomining	Plants absorb metal compounds; burned to ash
2	Bioleaching	Bacteria produce leachate containing metal compounds
3	Why new extraction methods are needed	High-grade ores are running out
4	How copper is got from its compounds	Displacement with scrap iron, or electrolysis
5	Life cycle assessment stages	Raw materials, manufacture, use, disposal
6	Why LCAs can be biased	Some values are judgements
7	Benefit of reducing, reusing and recycling	Saves resources, energy and waste
8	How glass bottles are recycled	Crushed and melted
9	How metals are recycled	Melted and recast
10	Why recycling metals saves energy	Less mining and extraction

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AQA GCSE Chemistry • Set 3: Corrosion and Alloys Paper 2 Triple only

#	Question	Answer
1	Corrosion	Destruction of a material by reaction with its environment
2	What rusting needs	Air (oxygen) and water
3	Barrier methods	Paint, grease, plastic coating, electroplating
4	Sacrificial protection	More reactive metal corrodes instead
5	Galvanising	Coating with zinc
6	Bronze	Copper + tin
7	Brass	Copper + zinc
8	Purity of 24 carat gold	100%
9	Steel	Iron + carbon (and other metals)
10	Stainless steel	Contains chromium and nickel; resists corrosion

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AQA GCSE Chemistry • Set 4: Ceramics, Polymers and Composites Paper 2 Triple only

#	Question	Answer
1	Soda-lime glass is made from	Sand, sodium carbonate, limestone
2	Borosilicate glass is made from	Sand + boron trioxide
3	How clay ceramics are made	Shape wet clay, heat in a furnace
4	Why LD and HD poly(ethene) differ	Different catalysts and conditions
5	Thermosoftening polymers	Soften when heated (no cross-links)
6	Thermosetting polymers	Do not melt (cross-links)
7	Composite	Reinforcement held in a matrix (binder)
8	Examples of composites	Fibreglass, carbon fibre, concrete
9	High carbon steel	Strong but brittle
10	Use of aluminium alloys	Aircraft (low density)

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AQA GCSE Chemistry • Set 5: The Haber Process and Fertilisers Paper 2 Triple only

#	Question	Answer
1	Haber process makes	Ammonia
2	Raw materials	Nitrogen (air), hydrogen (natural gas)
3	Equation	$N_2 + 3H_2 \rightleftharpoons 2NH_3$
4	Conditions	450 °C, 200 atm, iron catalyst
5	How ammonia is removed	Cooled so it liquefies
6	What happens to unreacted gases	Recycled
7	Elements in NPK fertilisers	Nitrogen, phosphorus, potassium
8	Source of potassium	Potassium chloride and sulfate (mined)
9	Phosphate rock is treated with	Nitric, sulfuric or phosphoric acid
10	Ammonia + nitric acid	Ammonium nitrate